

SE LAB - 1

Vihaan Wadhwa

PES1UG24AM319

Sec-F

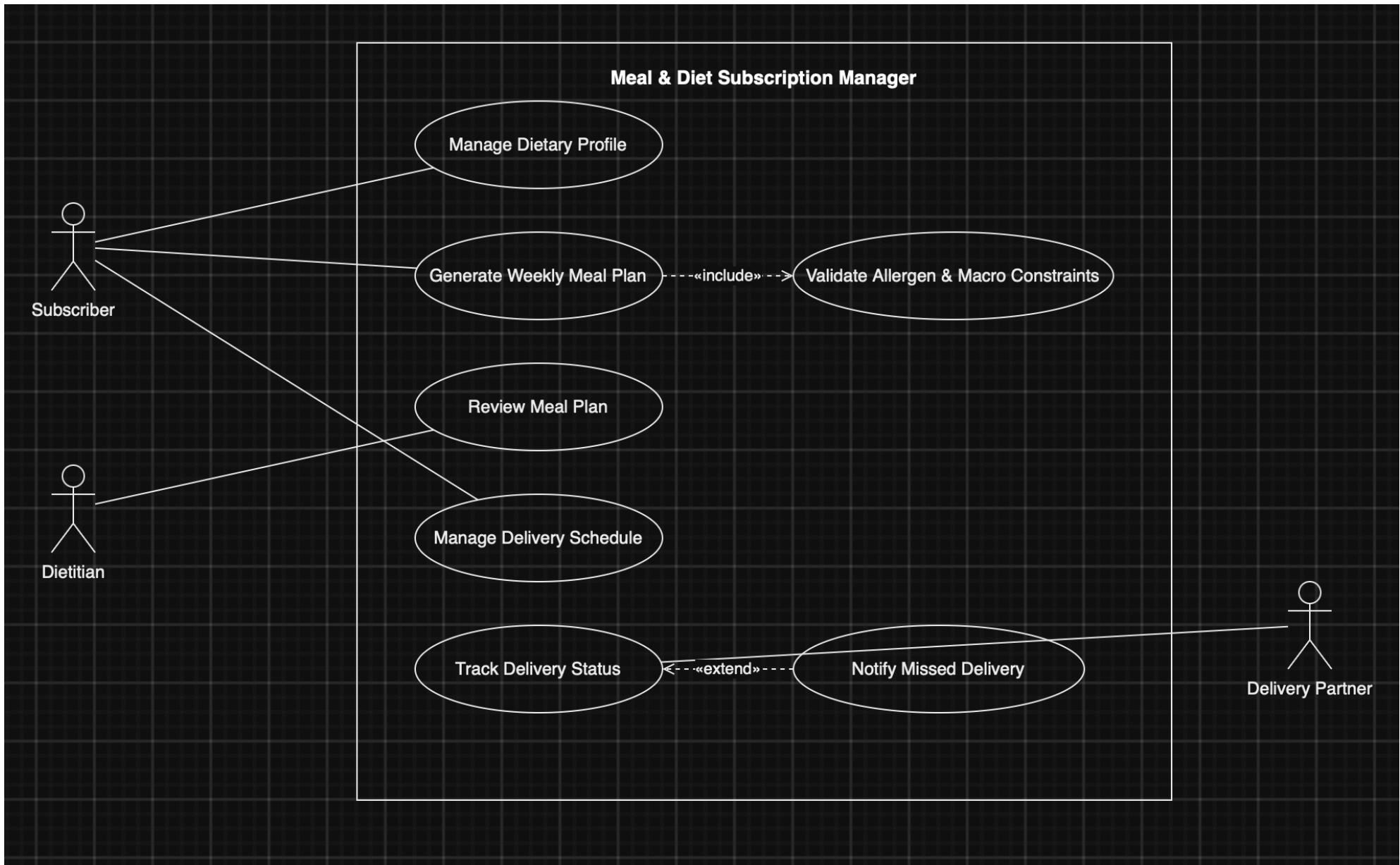
Problem Statement - 19

Personalized Meal & Diet Subscription Manager

1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall generate weekly meal plans strictly adhering to user macro-nutrient constraints and allergen exclusion lists.	High	Pass: Generated menu contains zero excluded allergen ingredients and follows the user's specified macro-nutrient constraints. Fail: Any excluded allergen is included or a specified macro-nutrient constraint is violated.	Core functionality of personalized meal planning and patient safety.
FR-002	Functional	The system shall allow the subscriber to create and update their dietary profile, including allergies, diabetic limits, calorie goals, and macro-nutrient constraints.	High	Pass: All entered dietary constraints are saved and displayed correctly when the profile is reopened. Fail: Any entered constraint is missing or incorrect.	Accurate dietary information is required to generate personalized meal plans.
FR-003	Functional	The system shall allow the subscriber to pause, skip, or reschedule an upcoming meal delivery.	High	Pass: A valid pause, skip, or reschedule request is recorded and the updated schedule is displayed. Fail: The requested modification is not recorded or displayed.	Allows subscribers to manage their meal subscription according to their needs.
FR-004	Functional	The system shall record and display the fulfillment status of each scheduled daily meal delivery.	High	Pass: When a delivery fulfillment status is updated, the corresponding delivery record displays the updated status. Fail: The delivery record does not reflect the updated status.	The scenario explicitly requires tracking of daily delivery fulfillment.
FR-005	Functional	The system shall allow the dietitian to review a subscriber's dietary constraints and generated weekly meal plan.	Medium	Pass: When a dietitian selects a subscriber, the system displays the subscriber's current dietary constraints and weekly meal plan. Fail: Either the constraints or meal plan cannot be displayed.	Allows the dietitian to review the information used for personalized meal planning.
NFR-001	Nonfunctional	The subscription engine shall support schedule modifications (pause, skip, reschedule) up to 12 hours before dispatch.	High	Pass: A pause, skip, or reschedule request submitted at least 12 hours before dispatch is accepted. Fail: A valid request submitted at least 12 hours before dispatch is rejected.	Ensures subscribers can modify their schedule within the specified time window.
NFR-002	Nonfunctional	The system shall restrict subscriber dietary and meal-plan information to authenticated and authorized users.	High	Pass: An unauthenticated or unauthorized user attempting to access subscriber dietary information is denied access. Fail: The information is displayed to an unauthorized user.	Protects sensitive subscriber dietary information in a healthcare-related system.

2. UML Use-Case Diagram



3. Use-Case Flow Document

Use Case Name: Generate Weekly Meal Plan

Primary Actor: Subscriber

Preconditions

1. The subscriber has a dietary profile in the system.
2. The subscriber's allergies, diabetic limits, calorie goals, and macro-nutrient constraints are available.
3. The subscriber has an active meal subscription.

Postconditions

1. A valid weekly meal plan is generated.
2. The meal plan complies with the subscriber's dietary constraints and allergen exclusions.
3. The subscriber can view the generated weekly meal plan.
4. The meal plan is available for delivery planning.

Main Success Scenario

1. The subscriber selects the option to generate or view the weekly meal plan.
2. The system retrieves the subscriber's dietary profile and stored constraints.
3. The system validates allergen exclusions, diabetic limits, calorie goals, and macro-nutrient constraints.
4. The system generates a weekly meal plan based on the validated constraints.
5. The system checks the generated meal plan against the subscriber's allergen exclusions and nutritional constraints.
6. The system confirms that the meal plan meets all validated constraints.
7. The system displays the generated weekly meal plan to the subscriber.
8. The subscriber reviews the weekly meal plan.
9. The use case ends successfully.

Alternate Flow — Meal Plan Fails Validation

1. The system detects that the generated meal plan contains an excluded allergen or violates a dietary constraint.
2. The system rejects the generated meal plan.
3. The system regenerates the weekly meal plan using the validated constraints.
4. The system validates the regenerated meal plan.
5. If validation succeeds, the system displays the compliant weekly meal plan.
6. If validation continues to fail, the system informs the subscriber that a valid meal plan could not be generated.